AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T318
Applied Network Security

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6/Dec. /2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2h

Instructions:

- Total Marks: **60**
- This exam consists of **6** Questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

AOU EGYPT

CamScanner

Question (1) Choose the correct answer (10 Marks): -

1. Which of the following is a key characteristic of classical encryption techniques?
   a. Use of asymmetric keys
   b. Lack of complexity in algorithms
   c. Use of substitution and transposition methods
   d. Dependence on modern computational power
2. Which of the following best distinguishes polyalphabetic ciphers from classical monoalphabetic substitution ciphers?
   a. Use of a single key for all letters
   b. Fixed substitution for each plaintext letter regardless of position
   c. Variable substitution based on a repeating key stream, flattening frequency analysis
   d. Reliance on transposition grids for diffusion
3. In number theory, for a modulus n = pq where p and q are distinct primes, what is φ(n), Euler's totient function?
   a. $(p-1)(q-1)$
   b. $p * q - 1$
   c. $(p+q)/2$
   d. $The\ number\ of\ integers\ coprime\ to\ n\ up\ to\ n-1$
4. The DES algorithm uses how many rounds of Feistel processing, and what is the effective key length after permutation?
   a. 16 rounds, 56 bits
   b. 12 rounds, 64 bits
   c. 20 rounds, 48 bits
   d. 8 rounds, 128 bits
5. Which of the following block cipher modes is most vulnerable to pattern replay attacks without proper IV management?
   a. Cipher Block Chaining (CBC)
   b. Galois/Counter Mode (GCM)
   c. Output Feedback (OFB)
   d. Electronic Codebook (ECB)
6. In linear feedback shift registers (LFSRs) used for stream ciphers, what property ensures maximal period length?
   a. Primitive polynomial over GF(2)
   b. Even number of taps
   c. Feedback from the least significant bit only
   d. Fixed initial state of all zeros
7. Beyond the CIA triad, what additional principle addresses the origin and authenticity of messages in network security?
   a. Non-repudiation
   b. Availability scaling
   c. Integrity hashing
   d. Confidentiality zoning
8. True random number generators (TRNGs) for cryptographic keys primarily rely on what to achieve high entropy?
   a. Pseudorandom algorithms like Mersenne Twister
   b. Physical phenomena such as thermal noise or radioactive decay
   c. User keystroke timings alone
   d. Deterministic clock cycles

EGYPT

CamScanner

9. In the Diffie-Hellman key exchange, what vulnerability arises if the discrete logarithm problem is solvable efficiently?
   a. Man-in-the-middle attacks become trivial
   b. The shared secret can be computed from public parameters
   c. Public keys are no longer verifiable
   d. Replay attacks are amplified
10. What is the avalanche effect in block ciphers, and why is it critical for security?
   a. A single bit change in plaintext causes total output inversion; ensures diffusion
   b. Key expansion leading to round variations
   c. Linear approximations in S-boxes
   d. Error propagation in modes like CBC

Question (2) Answer the following questions (10 Marks): -

1. What is the role of pseudorandom number generators (PRNGs) in modern cryptographic protocols, and why are they preferred over TRNGs for high-volume use? (2 Marks)
2. What is a transposition cipher, and how does it differ from substitution in terms of cryptanalytic resistance? (2 Marks)
3. Explain the purpose of the MixColumns transformation in AES and its contribution to diffusion. (2 Marks)
4. Encrypt the plaintext "SEND MESSAGE" using a Row/Column Transposition with key 312 . Provide the ciphertext.(4 Marks)

Question (3) Answer the following questions (10 Marks): -

1. Explain the concept of diffusion in cryptography and how it is achieved in AES. (2 Marks)
2. What role does the key expansion algorithm play in AES, and why is it designed to be non-reversible? (2 Marks)
3. Explain the main components of the Feistel structure and how they contribute to its security. Additionally, discuss why the number of rounds is crucial for the effectiveness of this structure. (2 Mark)
4. What is the function of the SubBytes step in AES, and how does it introduce non-linearity? (2 marks)
5. Consider the following ciphertexts. [seyosxisstuecixhra- cieaexshxettrbxass-fsxbosrrlteweixuco- sshxktoxeisxeorxdyot] output from a rows/column's transposition cipher using the key [463152 -164325- 236451 - 53124]. What is the plaintext for each cipher?[4 Marks]

Question (4) Answer the following questions (10 Marks): -

1. Describe the significance of using the nonce in cryptographic protocols (2 Marks).
2. Explain how a block cipher modes of operation like CBC impact security? (2 Marks)
3. What is a timing attack, and how can it compromise cryptographic security? (2 Marks)
4. Given an encrypted message "EVHQO", decrypt it using a Vigenère cipher with the keyword "KEY" (4 Marks).

Question (5) Answer the following questions (10 Marks): -

1. Calculate the totient function $\varphi(105)$ (2 Marks).
2. Using the extended Euclidean algorithm, calculate the multiplicative inverse of 97 modulo 312 (2 Marks).

CamScanner

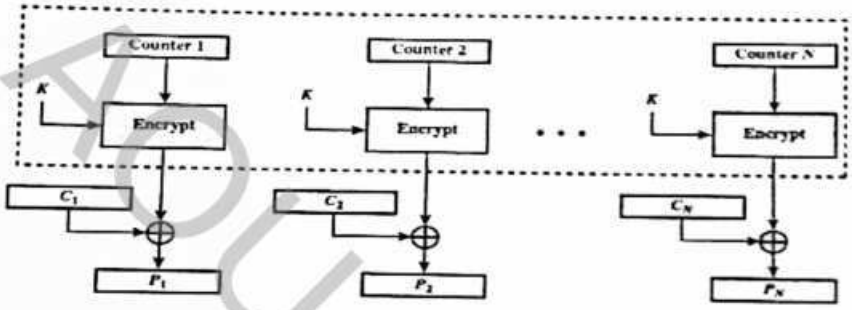
3. Explain What criteria should be used to select an encryption algorithm for secure communications? (2 Marks)
4. Assume when encrypting 3-bit plaintext with a block cipher with key K, the following ciphertext is obtained:

P	C
000	110
001	001
010	111
011	010

P	C
100	000
101	011
110	101
111	100

The following four ciphertexts was encrypted with the above cipher with key K in counter mode (initial value 0): What is the plaintext?(4 Marks)

a. 00110000
b. 101111010
c. 010110001
d. 00101010



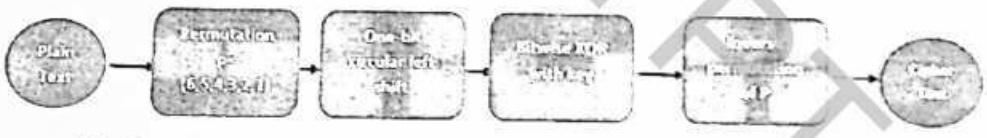
(b) Decryption

Counter Mode of Operation

Question (6) Answer the following questions (10 Marks): -

1. What is a ciphertext-only attack?(2 Marks)
2. Explain the principle of Kerckhoffs's principle in cryptography. (2 Marks)
3. Explain the role does entropy play in cryptographic key generation? (2 Marks)

1. A plaintext bit string 1 1 0 0 1 1 is converted into the ciphertext using the following sequence of operations. If the key is 0 1 0 0 1 0, what is the ciphertext? (4 Marks)



Note: consider Permutations and inverse permutation matrices are the same.

End of Questions

ABU-BYAT

CamScanner